

Designing efficient shared mobility services directly adapted to the spatio-temporal features of the potential demand

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1 Introduction

This paper presents a new method to precisely estimate the potential demand of shared mobility services. This method provides an aggregated vision of the spatial and temporal distribution of the demand. Contrary to the existing papers, this study only focuses on the demand without assuming any specific shared mobility services. This method is therefore generic and adaptable to any kind of service. The main objective of this work is to detect in a city a set of spatio-temporal areas in which the demand is high and recurrent over the time. Detecting such groups constitutes a considerable help for Transportation Network Companies to better understand the distribution of the demand and so to improve the proposed mobility solutions. Our method was experienced on a data set provided by the NYC Taxi and Limousine Commission. It give access to the origin and destination locations, but also the departure and arrival times of trips realized by their customers for 30 observation days. Even if this dataset is not fully representative of the real demand, it constitutes a good proxy of the mobility of the city. We focused our study on two districts of Manhattan well known in the literature as high density area in terms of mobility : Midtown and Upper East Side.

The rest of the paper is organized as follows : Section 2 presents the method of detection of similar trips, regular demand patterns. Then, Section 3.3 presents the results obtained, and introduces a method to check that regular patterns that have been identified in the historical data can be used to predict the demand for a new day. Finally, Section 4 is devoted to a final discussion.

2 Method

All the simultaneous shared mobility services, i.e. services where two passengers share the same vehicle to accomplish their trip, rely on the crucial assumption that, in a city, it exists travelers who perform similar trips. Hence, the paper is devoted to this central question is: how many users perform similar trips in a city and where are they located? Earlier studies only provide partial and incomplete answers. On the one hand, the largest part of the existing literature considers a specific service and identifies the associated demand to optimize the performance of the service [1, 2]. Undoubtedly, these studies provide valuable insights and inestimable contributions to the community either on the fleet size reduction, to improve performance efficiency of urban taxi service system [3], or to identify shareability characteristics of transportation network [4]. However, these approaches remain driven by one specific shared mobility service preventing them to identify and estimate the global demand for any kind of services. On the other hand, the existing studies on the analysis of ridesourcing or ridesplitting trips using observed data are quite limited. Especially, the first research to report an empirical analysis of ridesplitting, and the most significant for our work, is [5]. The authors develop a method to identify ridesplitting trips into TNCs data. Characteristics of such trips are compared to single ride and show that only 7% of the trips are shared. The main reason of this low percentage may be explained by the extra delay (10 min on average) and detour (1.55 km on average) imposed. Unfortunately, the authors only focus on past ridesourced trips and do not try to extract which single rides could have been shared by travelers. In other words, estimating demand by only considering spatio-temporal features of individual trips may lead to promising results and insightful contribution for the research community. Nowadays, such information is easily accessible through different sensing technologies: GPS trajectories, CDR data, etc. In particular, TNCs provide large data sets of pickups (PU) and drop-offs (DO) locations and times of trips performed by their drivers. Even if these trips correspond to a specific category of travelers and are not fully representative of the global flows of the city, these data sets can provide a unique insight on the shared mobility services demand. Moreover, the proposed methodology is not idiosyncratic and can be directly adapted to other sources of trips data. In this paper, we aim to pursue investigations of the shared mobility services demand and derive mobility patterns but by focusing on the characteristics of individual trips without assuming any specificity on the operation of the services. Our method is based on two main components :

a similarity function to estimate the likeness between pairs of trips and a clustering method to group similar trips. After calculating the similarity index for each pair of trips, we build a similarity matrix and give it as a parameter of the clustering algorithm. This method allows to detect groups of similar trips (based on the similarity index) with different sizes and characteristics.

2.1 Detection of similar trips

Before gathering similar trips, we have to define a similarity function in order to estimate the similarity between two trips. We describe in this section a new function of similarity based on the spatio-temporal proximity between departures and respectively arrivals of two trips. The observations used to test the proposed method are provided by the NYC Taxi and Limousine Commission. We focused our study on Midtown and Upper East Side on the period from 8am to 11am. This area has already been described in the literature as a very high density area [6].

2.1.1 Function of Similarity

Functions of similarity based on the spatio-temporal difference between departure and respectively arrival points of trips have already been studied in the literature : [7]. Nevertheless the main limitation of this approach to evaluate the similarity (such as the other attempts based on the Euclidean distance) is that it does not discriminate and enhance sufficiently the difference between trips, when they are applied to origin / destination locations and desired departure / arrival times only. Let i, j two trips and $l \in [PU, DO]$. If $l = PU$ then $d(p_i^l, p_j^l)$ represents the spatial difference between the departure points. Else if $l = DO$ then $d(p_i^l, p_j^l)$ represents the spatial difference between the arrival points. In the same way, $|t_i^l - t_j^l|$ represents the temporal difference between the departure times (respectively arrival times). A concrete example of this limit appears when we calculate the similarity of two trips with similar origin / destination, i.e. $d(p_i^l, p_j^l)$ close to zero. But with significant difference in their desired departure / arrival times, i.e. high values of $|t_i^l - t_j^l|$, the similarity index remains still low. As a consequence, it really deteriorates the ability of clustering methods to capture similar trips.

To overcome this drawback, we define a function $S(i, j)$ to evaluate the similarity between two trips i and j based on the characteristics of their origin and destination [8]. From a physical point of view, the intuition is that two (or more) travelers may have an interest to share their trip if they start in the same neighborhood and at the same moment, and want to go to the same destination. The function S must encompass these different spatio-temporal attributes of the trips. To fill this lack of modeling, we proposed the following function:

$$\overline{S(i, j)} = \sum_{l \in [PU, DO]} \alpha_l e^{|f^l(i, j)|} \quad (1)$$

where $f^l(i, j)$ is the feasibility function and α_l is a coefficient. Function f describes the service's potential to operate the shared trips, i.e. the ability to pick up (or drop off) the two travelers before both desired departure times:

$$f^l(i, j) = |t_i^l - t_j^l| - \gamma d(p_i^l, p_j^l) \quad (2)$$

where γ is the average pace to connect travelers that want to share a trip. This parameter is a general and synthetic formula to describe the operation of the service and the way in which this service gathers two demand requests into the same vehicle: defining a meeting point, successive pick-ups, etc. For example, if the first traveler must walk to the second traveler's pick-up point, then γ is equal to the inverse of the walking speed. If this distance is traveled by car, meaning that the service offers door-to-door service, then γ is equal to the inverse of the vehicle speed. Consequently, f is positive if the match can be realized before the two desired departure times t_i^l and t_j^l , whereas f is negative if travelers have to experience delay to make the match possible. γ is fixed exogenously (by testing different values) and macroscopically, i.e we use the same value for all the trips. Moreover, α_l is equal to $\frac{1}{2}$ if $f^l(i, j) > 0$ and to $\frac{3}{2}$ otherwise because it is more disadvantageous to be delayed.

In addition to this first index of similarity $\overline{S(i, j)}$, excessive distances / durations for rendezvous are penalized. Thus, penalties θ_x^l and θ_t^l are added when, respectively, the distances between origin (or destination) locations and departure (or arrival) times of trips i and j exceed, respectively, specific thresholds, δ_x^l and δ_t^l :

$$\theta_x^l = e^{d(p_i^l, p_j^l) - \delta_x^l} \quad \forall l / d(p_i^l, p_j^l) > \delta_x^l \quad (3)$$

$$\theta_t^l = e^{|t_i^l - t_j^l| \cdot \frac{\delta_t^l}{\delta_t^l} - \delta_t^l} \quad \forall l / |t_i^l, t_j^l| > \delta_t^l \quad (4)$$

Otherwise, these penalties are null. In this manner, $S(i, j) = \overline{S(i, j)} + \theta_x^l + \theta_t^l$ defines a function that enhances the differences between trips and facilitates identification of similar travelers in the data set. Notice that S is minimal (and equal to 1) when the two trips are exactly identical. This means that it will ease to cluster similar trips with a good intra-cluster homogeneity and, simultaneously, a significant inter-cluster dissimilarity.

2.1.2 Clustering method

Now, we can identify groups of travelers that may have an incentive to share a vehicle, according to the spatio-temporal features of their trips. To this end, trips are gathered using a clustering method. Among the large body of existing work, the main significant contributions for our study are the density based methods. Indeed, we want to determine spatio-temporal areas where similar trips occur. Moreover, it is important to use an algorithm able to detect clusters of different densities, which is not the case for all density-based algorithms. Actually, shared mobility services demand can take several aspects and intensity. Between the different choices, DB-SCAN holds our attention. This famous and classical method only requires two parameters : a threshold ϵ and a minimum number of points $MinPts$ which have to be in a radius ϵ so that the studied point is considered as an element of the cluster, see [9] for more details. Parameter ϵ is the maximal distance between trips, i.e. maximal value of S , allowed to consider them as similar and group them into the same cluster. However, this method must be slightly adapted to detect groups of different density. We decide to perform successive DB-SCAN clustering, using the similarity function S as the distance, while updating iteratively the values of the parameters. Starting with a large value of $MinPts = M$ and a drastic ϵ , it makes it possible to identify large groups of travelers in the initial data set of trip T . In other terms, we first detect large and high density clusters. Then, the DB-SCAN method is applied on the remaining non clustered trips to detect groups of size $M - 1$. This process is repeated until $MinPts = 2$. Simultaneously, we have defined an indicator of quality to select only the most promising clusters. A cluster can be considered satisfactory if locations of origin / destination and arrival / departure times of the trips within the cluster are relatively close. To this end, the function S is extended to consider sets of trips. In other terms, $|t_i^l - t_j^l|$ and $d(p_i^l, p_j^l)$ are respectively replaced by the mean distances, i.e. $\frac{1}{n_k} \sum_{i=1}^{n_k} |t_i^l - t_j^l|$ and $\frac{1}{n_k} \sum_{i=1}^{n_k} d(p_i^l, p_j^l)$ where n_k is the number of trips inside the cluster k . We then normalized these values because the acceptable delays are strongly related to the length of the trips. Consequently, the quality index of cluster k , $Q(k)$, is the function S applied to the set of clustered trips divided by the average length of the trips within cluster k . Notice that we aim at minimizing the quality index, i.e. best clusters present values of Q close to zero. Indeed, $Q(k)$ is low when (i) the spatio-temporal distance between origins is low, (ii) the spatio-temporal distance between destinations is low, and (iii) the mean travel distance is large. It comes that a cluster k is selected if and only if $Q(k)$ is below a specific threshold Q_{max} . In the remaining of the study, we set Q_{max} at 3. It corresponds to a restrictive matching policy. An example of clusters with different characteristics is depicted on Figure 1. Green markers denote departure points and red markers denote arrival points. We observe that the clusters can contain a number of trips greater or lesser, but the departure points and respectively arrival points are always spatially and temporally close to each other. This result proves that our similarity function is relevant. Moreover we can adjust the thresholds of our function or the DB-SCAN parameters to obtain clusters of different densities.

2.2 Detection of regular patterns

Because characteristics of the clusters do not depend on the observation days, it is thus appealing to investigate if commonalities in the clusters of the 30 days of the data set can be identified. Many approaches exist to derive the most representative partition from a group of partitions such as meta-clustering or consensus clustering method [10]. Here, we use the same clustering method, to maintain consistency when scaling-up. Thus, for each of the clusters identified for the different days of the data set, centroids are determined. Based on the characteristics of these centroids, i.e. mean origin / destination locations and mean departure / arrival times, indexes of similarity S between any pairs of centroids are calculated. Then, our algorithm can be performed to obtain groups of clusters. It makes it possible to identify if recurrent patterns exist in the shared mobility transportation demand but also to characterize these patterns based on their features: spatio-temporal localization, number of clustered trips, etc. For the sake of simplicity, we only focus on 19.466 clusters of 14 observation days, from 8 am to 11 am. The results are presented in the section 3.

3 Results

3.1 Number of clustered trips per day

Figure 2 now presents the clustering results for the 30 days of the data set. On average our algorithm detects 1410 clusters per day. Moreover we observe that the percentage of clustered trips per day is close to 85% for each day of our data set. This suggests to us that our method is very robust since the parameters we used do not change with days. It also means that the number of cars flowing in the network can be significantly reduced. Even if the Taxi data is not fully representative of the complete traffic flow, these results highlight that shared mobility could be a promising strategy to improve the performance of the transportation system by reducing the number of vehicles.

3.2 Number of recurrent clusters

With our method, 2136 meta-clusters, i.e. regular patterns among the days of the historical dataset can be obtained. It corresponds to a ratio of 94.81% of clusters that are recurrent from one day to the other. The

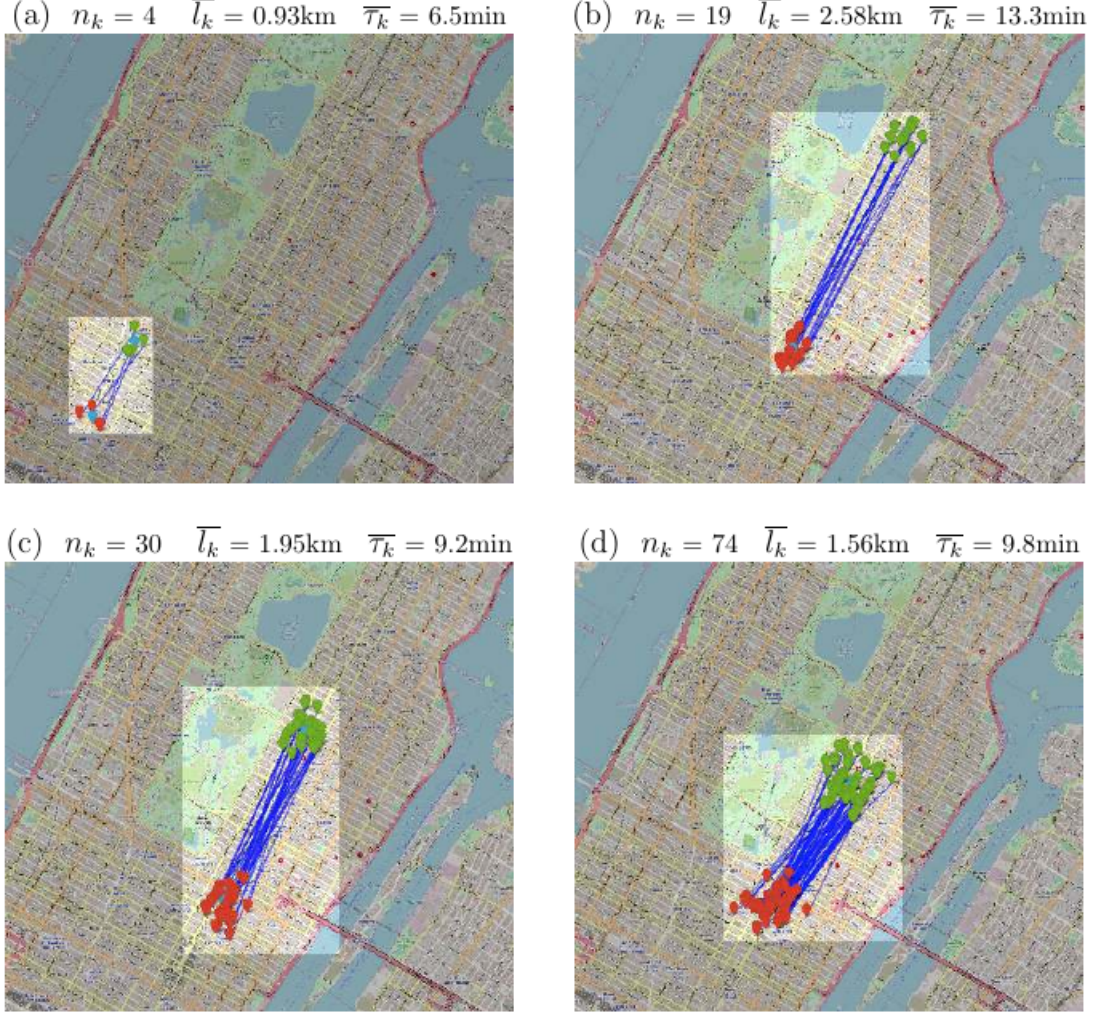


Figure 1: (a),(b),(c),(d) shows clusters of different sizes and characteristics. n_k denotes the number of trips in the cluster k , $\bar{l}_k$ denotes the average length of trips in k and $\bar{\tau}_k$ denotes the average duration of trips in k

detailed meta-clustering metrics are depicted in Table 1. The most interesting point is to observe the evolution of the trips forming a meta-cluster. To this end, Figure 3 simultaneously shows the evolution of the size of the daily cluster and the localization of the daily clustered trips, for a specific meta-cluster. It reveals that similar trips can be observed every day, from the same origin to the same destination at comparable desired departure and arrival times, except on week-ends (depicted by red dots in Figure 3). Similar patterns but with different characteristics can be observed for the other meta-clusters.

The characteristics of these meta-clusters, are very similar from those observed for the daily clusters. It also confirms the robustness of our demand estimation method.

3.3 Real-time application

We are now going to take advantage of these regular patterns to predict in a real-time how trips can be gathered into the same vehicle. First of all, it is important to remind that the graph of the demand can characterize different situations. Indeed, depending on the general trend of the demand curve of a meta cluster, it is possible to say whether the demand is regular, occasional or random. Figure 4 depicts several cases of curves of the demand. The blue curve shows a regular high demand, in this case it could be interesting to implement a regular bus line. The red curve depicts an occasional high demand, in this case we need to analyse more specifically why trips are performed only some days (football game, music concert etc.). The green curve correspond to a random demand, in this case it is very difficult to predict the number of trips for a specific future days. After performing the meta-clustering algorithm, we obtain a set of meta-clusters. This algorithm is very similar to the DB-SCAN that we used to detect groups of similar trips, but it works on the centroids of the clusters previously found

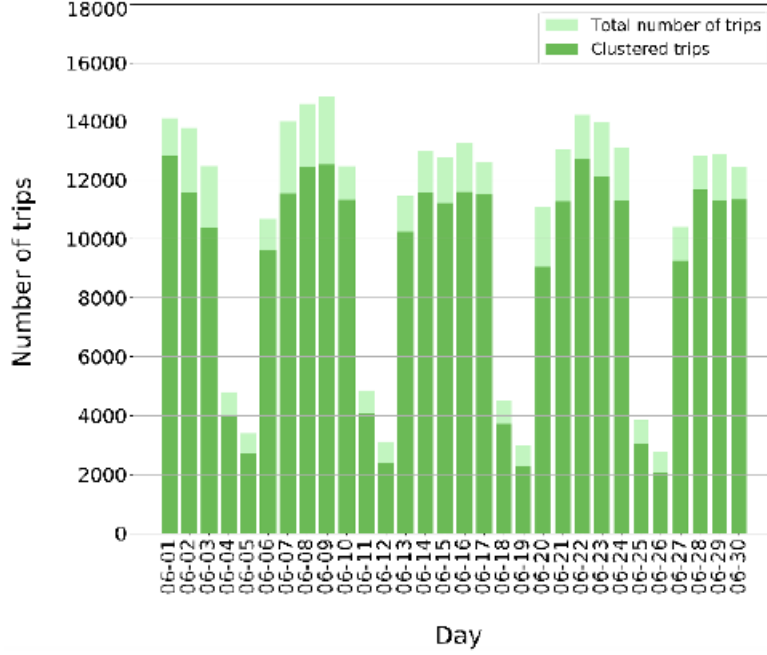


Figure 2: Clustered trips and total number of trips per day

Description	Value
Number of Meta-clusters	2136
Number of clusters grouped to form Meta-clusters	18456
Mean number of clusters by Meta-cluster	8.64
Mean distance between origins	0.33 km
Mean distance between destinations	0.33 km
Mean offset time between origins	9.5 min
Mean offset time between destinations	9.6 min
Mean distance of travels	1.71 km
Mean travels time	11.1 min

Table 1: Results of the meta-clustering carried out over the period from June 6 to 19, 2011

and not directly on the trips. Each meta-cluster corresponds to a spatio-temporal area in the city. So it seemed interesting to check if this pattern corresponded well to the demand for a new random day. In other words, we check for a new day the percentage of trips that will fall into a meta-cluster. Figure 5 shows that for a new day on average more than 99% of trips can be gathered into a meta-cluster. Thanks to this result we can say that the meta-clustering is a consistent method to predict the potential future demand of shared mobility. The authors' current researches revolve around the following question : is it possible to guarantee that there will be at least a number x of people making the same trip at the same time ? This work will be approached in a future paper.

4 Conclusion

In this paper, we investigated whether groups of travelers that realize similar trips in a city exist. These users constitute the potential demand of the future shared mobility services such as ridesharing, shared taxis or car-pooling. Obviously, the ability to share a trip with another traveler relies on many determinants such as the cost, the comfort, the reliability, etc. To tackle this challenge, only spatio-temporal features are considered here: origin and destination, desired departure and arrival times. Based on these attributes, a general methodology has been elaborated and experimented using data giving access to these information: set of trips realized by taxi users.

The methodology mainly relies on a function to evaluate the similarity between any pair of trips. This index drastically differs from the existing works because the shape of the function is very sharp (making it particularly convenient for clustering) and considers the feasibility of the trip sharing, i.e. ability of the transportation service to realize multiple pick-up / drop-off. Then, the classical DB-SCAN clustering method has been adapted to the

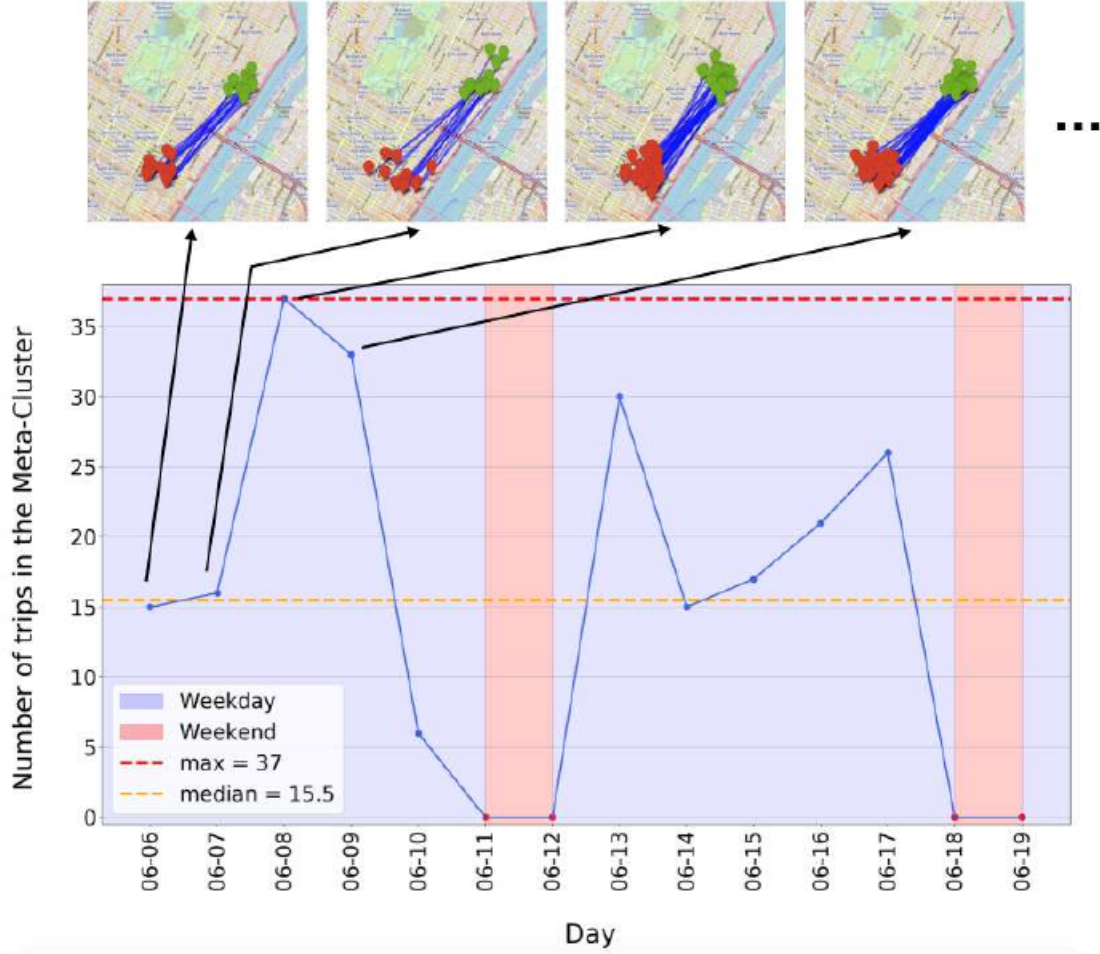


Figure 3: Analysis of the daily trips forming a meta-cluster: similar trips can be observed every day

specific case of the shared mobility services demand estimation. We have designed an algorithm able to gather similar trips into groups of different sizes and characteristics by fixing only the maximum dissimilarity allowed between trips of the same group. This method has then been experimented using a data set coming from NYC Taxi and Limousine Commission. However, the method is enough general to be performed with any other source of data giving access to origin / destination and desired departure / arrival times of trips : GPS trajectories, OD matrices, travel surveys, etc.

The results show that more than 85.5% of the trips can be clustered for the 30 observation days of the data set by maintaining a relatively restrictive matching policy. Travelers are gathered on groups of different sizes: from pairs of travelers to groups of 30 users that realize almost the same trips at the same time. The largest part of the clusters is composed of groups of 2 or 3 travelers, meaning that services such as ride-sharing, shared taxis, or car-pooling are the most promising solutions between all the different versions of shared mobility services. Then, we investigated if there were commonalities between the clusters of 14 successive days. Using the same clustering method, regular patterns have been identified among the daily clusters. Specifically, large clusters are recurrent, i.e. one can observe significant number of trips with same origin / destination and desired departure / arrival times from one day to the other. This really defines the recurrent demand of shared mobility services. Especially, these findings give valuable insights for future transportation supply providers to optimize their services: anticipation on the vehicle dispatch, deployment of virtual lines, identification of meeting points, etc.

Finally, the forecasting method showed that for a new random day, on average, more than 99% of trips fall into the meta-clusters previously established. This result confirms the idea that identifying regular patterns can be a reliable tool to predict future demand of shared mobility and directly provide the most appropriate transportation service.

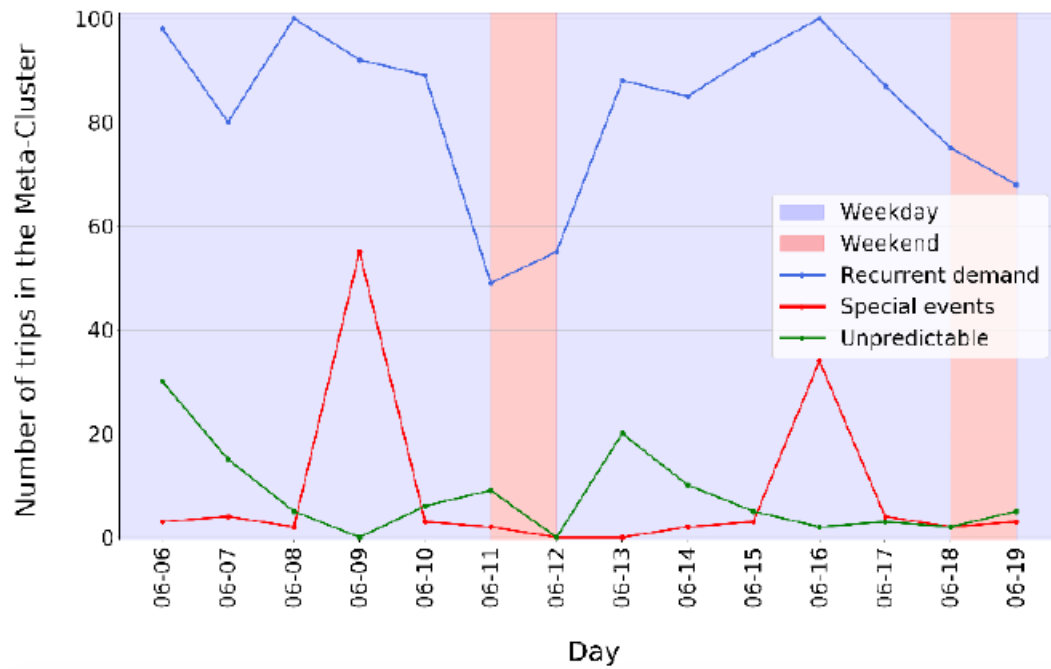


Figure 4: Different kind of demand graph.

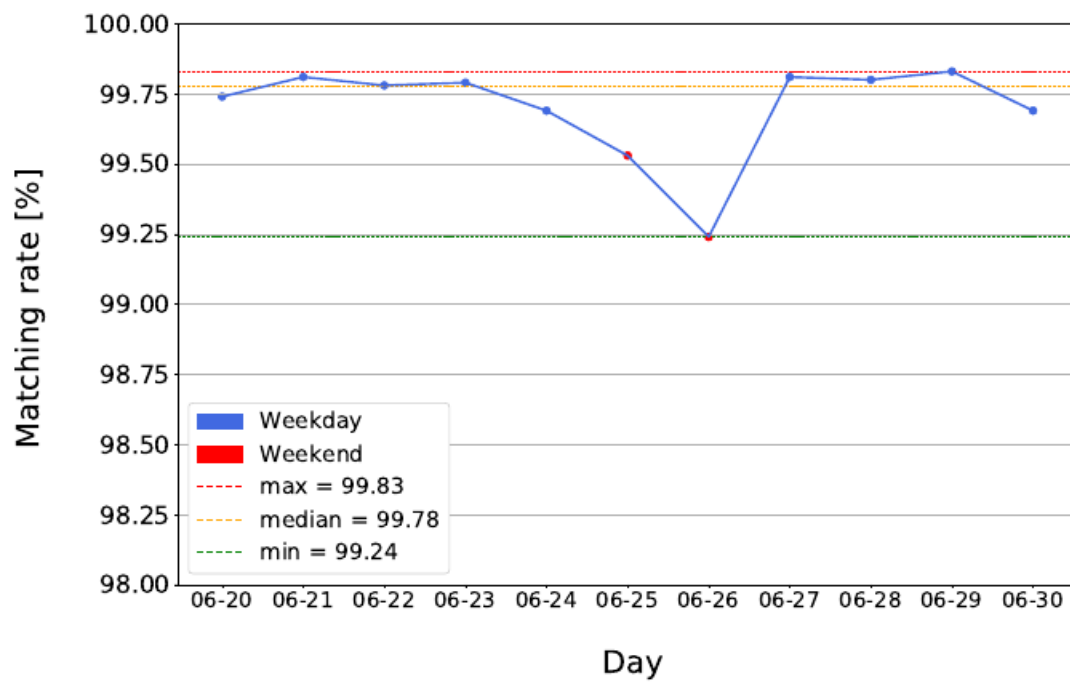


Figure 5: Matching rate of the forecasting method.

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